

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) Semester III (NEP) Examination 2024-25

BOTANY**Paper No. BOTMJ21 / BOTMN21 : General Microbiology**

Time: 2:30 Hours

Maximum Marks: 70

(Write your Roll No. at the top immediately on the receipt of this question paper)

Note: Attempt five questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

1. Answer all the following in not more than 50 words each: [7 × 2 = 14]

- (a) Who developed postulates to prove that a microorganism causes a specific disease?
- (b) What did Joseph Lister contribute to microbiology?
- (c) What are biodegradation and biotransformation?
- (d) What is active transport?
- (e) What is the bacterial capsule?
- (f) What is the biological source of the antibiotic tetracycline?
- (g) What are Halophiles?

2. A bacterial colony isolated and subsequently purified from the arid desert ecosystem of Rajasthan exhibited significant morphological heterogeneity upon detailed observation. What comprehensive set of parameters can be employed to accurately delineate its taxonomic identity and how can these approaches help in resolving the presence of multiple morphotypes within a single isolate? [14]

3. Write short notes on any two of the following: [2 × 7 = 14]

- (a) Write about phenotypic characteristics used to identify microorganisms.
- (b) In which scenario is DNA fingerprinting an effective tool for identifying microorganisms, and under what circumstances might it fail or be inadequate as a molecular marker?
- (c) What are xenobiotics? Discuss the role of microbes in their degradation with suitable examples.

4. What is the difference between T4 phage and Lambda (λ) bacteriophage? Describe the life cycles of these bacteriophages with appropriate diagrams. [14]

5. Answer any two of the following: [2 × 7 = 14]

- (a) Differentiate between the cell walls of Gram-positive and Gram-negative bacteria.
 - (b) Write about Endospores.
 - (c) Write about Thermophiles.
- 6.** How are antibiotics classified? Discuss the mechanisms involved in antibiotic resistance.
[14]
- 7.** Write short notes on any two of the following: [2 × 7 = 14]
- (a) Nucleoid
 - (b) Bacterial flagella
 - (c) Psychrophiles